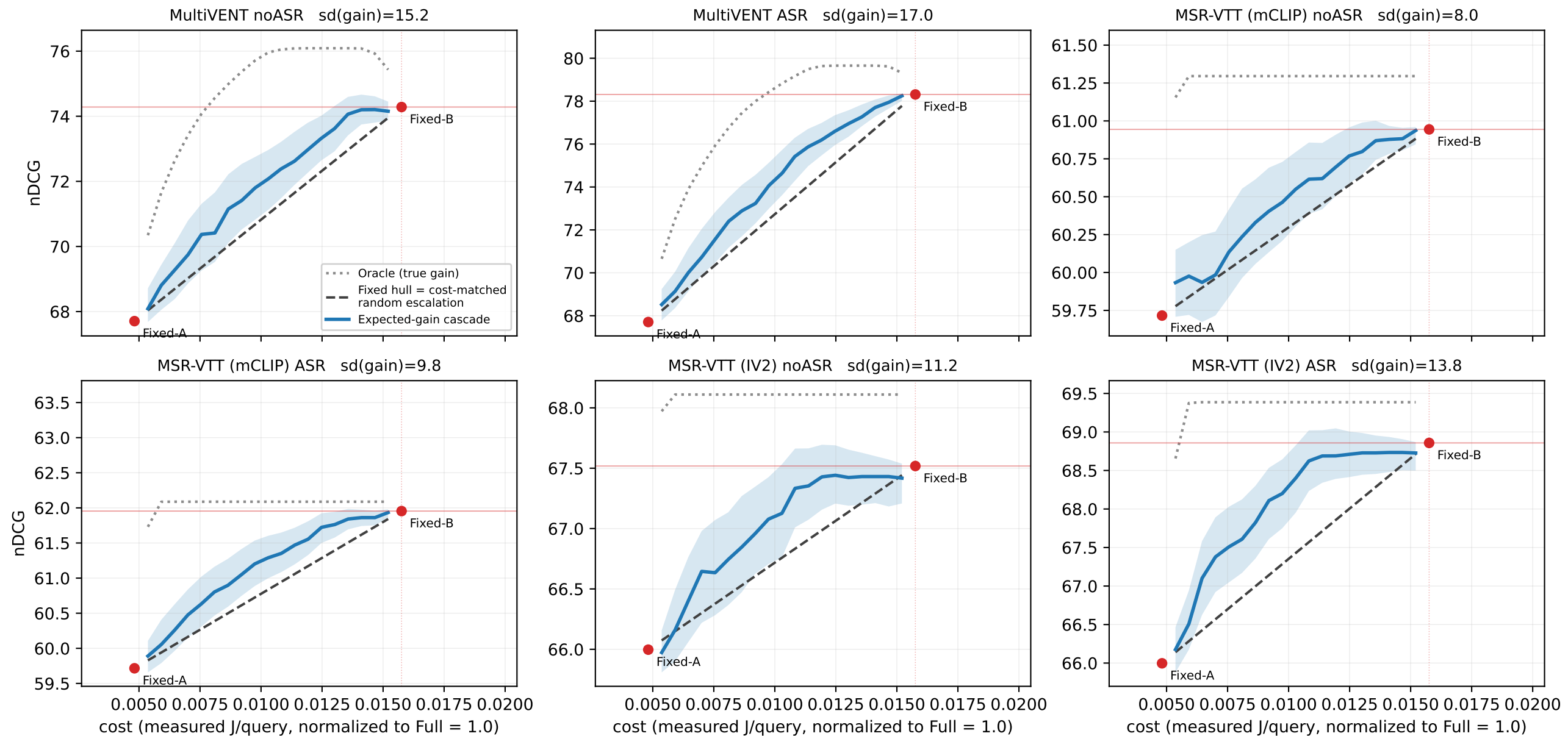


Per-query expected-gain routing beats the cost-matched fixed baseline in the low-budget regime
(the cascade never buys the Full tier: event-decomposition gain is not predictable per query)



x-axis is MEASURED energy: tier B = 1.58% of Full, tier A = 0.48% (Full = 1418 J/query, off-axis at 1.0). The component-count proxy overstated this range as 20-40% of Full.